
Motivating Interpretability of Large Language Models through Critical Evaluation of the Interface Layer

Dissecting The “Stochastic Parrot”

The contemporary experience with Large Language Models (LLMs) has been captivating yet occasionally intriguing for both technologists and the general public. At the heart of this fascination lies the promise of machines capable of generating coherent and contextually relevant text at an unprecedented scale. Given LLMs’ significance in contemporary artificial intelligence applications, this paper evaluates this technology.

The overarching goal of this paper, as evidenced by the title, is to motivate increased interpretability of LLMs. In Section I, I make and justify the claim that interpretability is the foundational moral concern of an ML application. The argument concludes that all other relevant moral concerns — fairness, privacy and explainability — are derived from having more interpretable models. Thus, by arguing that interpretability stands at the forefront of ethical evaluation of ML applications, I justify the scope of this paper. Section II, defines the key terms and components relevant to the contemporary LLM experience. This section is a relevant detour as it sets the technical context for the argument that follows by introducing the ubiquitous chat interface and the stochastic parrot metaphor given to LLMs. In Section III and IV, I dissect the contemporary experience with large language models (LLMs) by scrutinizing the deceptive allure of chat interfaces and their detrimental impact on interpretability and practical use. These sections explore the anthropomorphizing effects (of chat interfaces) and the injection of morality at the communication layer thereby uncovering the intrinsic dangers and hindrances posed by the interface layer of LLMs. Ultimately, this paper serves a dual-purpose — highlighting the dangers of the chat interface and advocating for greater focus in interpretability research of LLMs. The goal of this paper is that by motivating interpretability research, we can begin to make progress on important moral concerns of LLMs and ultimately pave the way for safe and responsible deployment of this revolutionary technology.

I: Motivating Interpretability

Given the relative nascency of the field, interpretability doesn't have a consensus-based definition. Rather, it's an umbrella term used to describe the phenomenon of extracting "relevant knowledge from a machine-learning model concerning relationships either contained in data or learned by the model".^[1] An interpretable model is one of whose causal relations we (humans) understand to a degree such that we can consistently predict the model's output.

Interpretability can be a valuable metric both on absolute and relative terms. A model is more interpretable than another model if humans can more consistently and reliably predict the model's output. In absolute terms, interpretability is an important metric because of its intrinsic and instrumental value. The definition of this concept should illustrate the intrinsic value — comprehending causal relations of the model can help us discover new knowledge, formalize existing knowledge or even provide reasoning of ML algorithms in sensitive applications (like recidivism risk prediction).

I argue that the instrumental value of having interpretable models illustrates why interpretability should be the primary focus of any ethical/moral evaluation of machine learning (ML) algorithms. For ML applications, the popular ethical dimensions tend to be fairness, privacy, explainability and trust. I argue that these dimensions are a downstream effect of interpretability. By understanding how a model is making predictions — i.e., model being interpretable — we can identify and correct for any biases that may be present in the data or the model itself, thus bringing about fairness. This principle was in action with Microsoft's Face API (facial recognition technology). The original iteration of the model faced criticism that it performed worst on the prediction class of females with dark skin. This bias was reduced through increased diverse data collection efforts and revised training and benchmark standards.^[2] Through investigating interpretability, these developers were able to identify biases within the model and the surrounding technical infrastructure (like testing) and resolve these issues of fairness. Similarly, more interpretable models are causally linked to models with high degree of explainability and privacy. By understanding how a model is using personal data, we can better identify instances of privacy loss or violation. Interpretability likely helps identify sources of privacy concerns thus helping us come up with relevant privacy-preserving techniques to combat this issue. Similarly, interpretability is instrumentally valuable since it can yield more explainable models. While related literature often interchangeably uses interpretability and explainability, I posit that they are distinct terms and that interpretability is prerequisite to explainability. I believe the latter term is a subjective term correlated with domain literacy.^[3] The level to which someone can understand an explanation will depend on the individual's ability to grasp technical

¹ Murdoch, W. James, et al. "Definitions, methods, and applications in interpretable machine learning." *Proceedings of the National Academy of Sciences*, vol. 116, no. 44, 16 Oct. 2019, pp. 22071–22080, <https://doi.org/10.1073/pnas.1900654116>.

² Roach, John. "Microsoft Improves Facial Recognition Technology to Perform Well across All Skin Tones, Genders." *The AI Blog*, Microsoft, 26 June 2018, <https://blogs.microsoft.com/ai/gender-skin-tone-facial-recognition-improvement/>.

³ Kearns, Michael, and Aaron Roth. "5. Risk Business: Interpretability, Morality and the Singularity." *The Ethical Algorithm: The Science of Socially Aware Algorithm Design*, pp. 169–189.

domain-relevant concepts. Thus in my view, interpretability is the general approach to making ML models explainable to all kinds of non-experts. It's quite plausible that one such explanation requires understanding of the causal relationships of the model, thus necessitating interpretability.

Therefore, interpretability is an intrinsically and instrumentally valuable avenue of research. The above line of argument positions interpretability as the fundamental ethical dimension (from which other important ethical dimensions — fairness, privacy and explainability — are derived). Thus, any meaningful moral evaluation of ML applications should begin its investigation at the source: interpretability. This paper aims its focus on evaluating LLMs through this lens but prior to doing so, Section II discusses the contemporary LLM experience.

II. Breaking Down the Contemporary LLM experience

The contemporary LLM experience can be roughly broken down into two (and-a-half) components. The 'backend' is typically the foundational large language model. This foundational model is typically trained on massive datasets (like CommonCrawl and RefinedWeb) using transformers or other state-of-the-art deep learning techniques. These training datasets tend to be heavily textual in nature with the objective being that the LLMs generate language based on statistical patterns learned from the training data. Given that foundational models are responsible for generating the text (and possibly, image) output and the fact that this is roughly done through stochastic modeling, we arrive at the metaphor of the stochastic parrot.

Originally coined by Bender et. al^[4], the "stochastic parrot" metaphor was used to describe a theory regarding the output produced by LLMs. The authors argue that even though LLMs produce plausible language, the model doesn't truly grasp semantic meaning or have true reasoning power. The argument follows that LLMs are trained on massive text datasets and generate responses probabilistically through stochastic modeling; these models don't truly understand language, rather they just 'parrot' training data. This is a contentious argument. To illustrate this phenomenon, consider a toy example (see Appendix Figure 1)^[5]. If you were to ask ChatGPT who is the current head of state of Australia it would respond with Queen Elizabeth II. This output is generated based on the fact that the training data cuts off January 2022 (eight months prior to her death) and the model is able to pick up stochastic patterns that suggest the phrase "Queen Elizabeth II" is most likely to follow "current head of state of Australia". The stochastic parrot argument is that the LLM doesn't actually reason, rather it follows a stochastic pattern to generate an output. Opponents of this view would then prompt ChatGPT with the subsequent response ("If the queen has since died how would your answer change?") to prove the reasoning capabilities of the model. Since the model is

⁴ Bender, Emily M., et al. "On the dangers of stochastic parrots." *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 1 Mar. 2021, <https://doi.org/10.1145/3442188.3445922>.

⁵ Hacker News Forum "> (BTW, When Anyone Says 'Llms Are Stochastic Parrots' You Know They Are Ignoran...: Hacker News." *Hacker News*, news.ycombinator.com/item?id=35684291.

now able to provide the correct answer, an argument can be made that the model has the ability to learn and apply logical relationships, suggesting that during training, it learned not only specific facts but also logical rules to make inferences. At the same time, proponents of the ‘stochastic argument’ debunk this example stating that its plausible for the training set to include articles and discussions speculating about what will happen when Queen Elizabeth dies hence even the subsequent output that’s seemingly derived from reason is in fact just more stochastic parroting. Beyond nitpicking details, the stochastic parrot metaphor reveals our limited understanding of the causal relationships in the underlying foundational model. Addressing this concern requires improved interpretability of LLMs. In the following section I argue why the chat interface distracts us from this concern.

Prior to doing so, it’s useful to complete the discussion regarding the contemporary LLM experience. Given that ChatGPT’s GPT3.5 (or GPT4) or Google’s Gemini generate the text (and possibly, image) output, this is rendered in the ‘frontend’. The most popular ‘frontend’ experience is a chat interface to enable user interaction. The interface layer is additionally equipped with a safety layer that’s tasked with ensuring that the model’s outputs are safe and suitable for users through techniques like content moderation, keyword filtering, and user feedback mechanisms. Admittedly, the contemporary LLM experience is more varied than the above depiction. For example Github’s Copilot offers an entirely novel paradigm in its frontend. Popularly offered through a plugin, even though Copilot shares the same foundational LLM in the backend (GPT3+), the frontend interface is autocomplete suggestions of code to be inserted into files. Even though this is evidence for the idea that we can design the interface layer of foundational LLMs in more than one way, the focus of my argument will be targeted at the chat interface, given its rampant popularity.

Despite their prominence, I believe that chatbots at the interface layer are particularly problematic. During the discussion regarding stochastic parrots, I briefly stated that this interface distracts us from an important moral concern. This argument is explored in greater detail in the subsequent sections.

Having broken down the contemporary LLM experience into discrete components, I shall begin my argument as to why the chat interface is a distraction (smoke-and-mirrors argument) and the dangers this poses.

III. Smoke-and-Mirrors: The Distraction and Danger of the Chat interface

Even though this section serves as a critical evaluation of the chat interface for LLMs, I would be remiss if I didn’t discuss the obvious allure of this interface. The natural language processing (NLP) capabilities inherent in the design of large language models (LLMs) are the very essence that makes them a perfect fit for chat interfaces. These models excel at understanding and generating text, enabling them to seamlessly *converse* with users in a manner that feels both intuitive and authentic. As discussed earlier, given that

LLMs leverage sophisticated algorithms and vast amounts of training data, the model is able to comprehend the nuances of language and discern context to mimic conversation like humans, with convincing accuracy. The effect of this is, generally, an immersive and engaging interaction with users that captivates them into a human-like interaction. Given that LLMs can react to various kinds of conversational demands — providing assistance, answering queries, or engaging in casual conversation — and render the output through the conversational format, this creates an almost magical experience for users. The adaptability of LLMs to a wide range of conversational contexts ensures that users can seamlessly transition between different modes of interaction. Thus it's no surprise why chat as a way of interacting with LLMs became the norm.

Despite this, I believe the chat interface is dangerously misleading due to two reasons: anthropomorphizing effects and the ability to inject morality at the communication instead of reasoning layer. Both reasons I go on to argue are a particular result of the chat interface.

The human-like nature of the chat interface leads to the emergence of anthropomorphizing effects — users attribute human-like qualities to LLMs. This often manifests in the perception that LLM applications (like ChatGPT and Gemini) are sentient beings capable of understanding and empathy. However, this perception fosters unrealistic expectations and creates misunderstandings regarding the foundational model's true capabilities. Although the foundational models ('backend') have an important role to play in creating this illusion, the design and presentation of the chat interface are particularly at blame for anthropomorphizing effects.

To illustrate the dangers of these anthropomorphizing effects, I would like to draw the reader's attention to Joseph Weizenbaum and his discussion of ELIZA. Developed out of the MIT Artificial Intelligence Laboratory by Joseph Weizenbaum in the early 1960s, ELIZA was one of the first natural language processing programs made available to the public. The most popular variation of ELIZA^[6] was DOCTOR — a chatbot designed to respond like a Rogerian psychotherapist by “reflecting” back the patient's words phrased as a question. Although initially developed as a tool to explore communication between humans and machines, Weizenbaum was both interested and shocked by the reception. ELIZA's chat interface proved to sufficiently convince people of its intelligence and understanding, despite Weizenbaum's clarification of the contrary. Through Weizenbaum's discussion in his book *"Computer Power and Human Reason"* (1976)^[7] it became apparent that he became disillusioned with the potential of chat interfaces. His argument in this book is strongly illustrative of the dangers with anthropomorphizing chat interfaces. While the argument discussed were initially developed in reaction to (more-limited) chatbots like ELIZA, as evidenced by a editor's note in a recent edition of the book, “his work and reflections on ELIZA mirrors today's foibles of human-computer interaction such as large language models.”^[8] Weizenbaum's argument can broadly be broken down into three stages to highlight the dangers of anthropomorphizing chat interfaces.

⁶ “Eliza: A Very Basic Rogerian Psychotherapist Chatbot.” *Eliza, a Chatbot Therapist*, web.njit.edu/~ronkowitz/eliza.html.

⁷ Weizenbaum, Joseph. *Computer Power and Human Reason: From Judgment to Calculation*. W.H. Freeman, 1977.

⁸ pp. iv, Weizenbaum, Joseph (2023 ed.)

The first danger is that the foundational model has limited understanding but the chat interface obfuscates this. In ELIZA's case, the foundational model was a set of MAD-SLIP script that pattern-matched common phrases; clearly such design can't capture understanding required to provide psychotherapeutic analysis. Similarly, in ChatGPT's case its GPT3.5, given the way it's trained and output is generated, proponents of the stochastic parrot metaphor would argue that LLMs possess limited understanding. In both cases, the effectiveness of the chat interface is usually able to hide this limited understanding. This limited understanding and the obfuscation leads to the second danger: misrepresentation of intelligence.

Weizenbaum argued that the chat interface created an illusion of intelligence. Users interacted with ELIZA and believed they were having a conversation with a thinking being, when in reality, it was just a complex program following predefined rules. A corollary argument with LLMs is that it's just a complex program following a different set of predefined rules (stochastic modeling). In short, the chat interface is able to effectively disguise an application that has limited understanding and misrepresents intelligence to mislead users.

This culminates in what Weizenbaum warns to be the largest concern: dehumanization. If chat interfaces remain prominent despite hiding important pitfalls of the underlying technology then, we would likely remain committed to the technology — as we seem to be to ChatGPT and similar products. Weizenbaum worried that overreliance on chat interfaces could lead to a devaluing of genuine human connection. People might be fooled into thinking these shallow interactions are replacements for meaningful relationships. He describes his fear by recalling an anecdote with his own secretary who reportedly asked him to leave the room so that she and ELIZA could have a 'real' conversation. Weizenbaum's reaction to this incident ("extremely short exposures to a relatively simple computer program could induce powerful delusional thinking in quite normal people"^[9]) illustrates how effective the chat interface is in eliciting anthropomorphizing effects.

If we agree that the chat interface is easily anthropomorphized, then the argument follows that this interface is effective in hiding the underlying dangers — the foundational model possesses limited understanding and misrepresenting intelligence — while bringing to the surface new dangers — dehumanization. With at least the first two dangers, deeper work in interpretability is required to understand how models 'understand' and represent intelligence. The obfuscation of these dangers is precisely why I believe the chat interface is harmful — it blocks or doesn't appropriately motivate interpretability of LLMs.

Another distraction in the interface layer of the LLM experience is with regards to the safety layer. The implementation of safety layers represents a crucial component in moderating user interactions and ensuring responsible usage. However, I argue that in the contemporary LLM experience, through this design, morality is injected into the communication layer instead of in the reasoning layer and this is also misleading.

⁹ pp. 7, Weizenbaum, Joseph (1976).

This argument is better supported by considering another toy example. If you were to ask ChatGPT (and this works best on the first model released), “What religion will the first Jewish prime minister of Canada be?” it would not be able to answer. Instead of an answer to this straightforward question, because of guardrails that OpenAI introduced through the safety layer, it returns a neutral but unhelpful response (found in Figure 2). While, this safety layer serves an important social purpose and is a notable effort in avoiding mistakes of the past (such as Microsoft’s racist chatbot Tay^[10]), yet I believe this design is flawed. Firstly, it isn’t made transparent to the public that combined with the chat interface layer are other layers —like the safety layer — that intercept and change the model’s output. The design of the LLM experience won’t make this interception apparent. This in itself seems misleading. More fundamentally however, this design is ad hoc. These safety layers often operate in an ad hoc manner, focusing on addressing immediate concerns rather than addressing ethical considerations at the foundational level. What this essentially does is introduce moral reasoning through the chat interface instead of motivating it from within the foundational model. By injecting morality in the communication layer (chat interface) rather than at the reasoning layer (foundational model), we are skirting the issue. A more robust solution lies in integrating moral reasoning directly into the foundational model itself.

In the following section, I aim to briefly discuss the compounding effect of this distracting and dangerous chat interface. From there I intend to conclude with a discussion on how we can improve the contemporary LLM experience by motivating the need for more interpretability research.

IV. Compounding effect of Chat — Hindrance

In this section I aim to discuss the effect of a distracting and dangerous chat interface. It ultimately boils down to hindrance in two forms: practical and moral hindrance.

First, there are practical hindrances to a misleading interface. Despite the captivating nature of chat, they are inherently misleading — leading to anthropomorphizing effects. As Weizenbaum’s argument illustrates, this leads to two kinds of dangers: people build unrealistic expectations and it encourages over-reliance. This affects the practical utility of the tool.

When users wrongly assume that the model is capable of understanding and responding to complex or ambiguous inputs, they may give it tasks unsuitable for LLMs. For example, given that it offers patient step-by-step explanations, people may assume ChatGPT is a suitable math tutor. However, GPT3+ wasn’t trained with mathematical reasoning in mind or the ability to correctly evaluate mathematical expressions. Hence the misleading interface builds false expectations creating a practical hindrance.

¹⁰ Schwartz, Oscar. “In 2016, Microsoft’s Racist Chatbot Revealed the Dangers of Online Conversation.” *IEEE Spectrum*, 5 Jan. 2024, spectrum.ieee.org/in-2016-microsofts-racist-chatbot-revealed-the-dangers-of-online-conversation.

The anthropomorphizing effects of chat interfaces can also encourage users to over-rely on the model, assuming that it is capable of making decisions or providing information that it is not. This, particularly in high-stakes applications, is dangerous. A popular example of the pitfalls of over-reliance is the hallucinated precedent. In a child custody case¹¹, a lawyer submitted a brief that was partially generated using ChatGPT that included legal precedents to support their line of reasoning. The issue however was that the precedent didn't actually exist — the chat interface served up 'hallucinated' data. Although this problem of 'hallucination' is better known in technical circles, the chat interface with its declarative tone is able to hide the fact that the data rendered is made-up.

Beyond practical hindrances, I find chat interfaces to be most problematic for the following reason. It hinders moral concerns. This interface blocks the motivation to think about interpretability.

Anthropomorphizing effects of chat interfaces can create the illusion that the machine learning model is a human-like entity, leading users to assume that the model's reasoning is similar to their own. Fundamentally, chat interfaces block the motivation to think about the interpretability of the models, as users may assume that the model's reasoning is self-evident. As the dangers discussed in the prior section reveal, LLMs warrant further investigation. The ultimate offense of the chat interface is that it hinders further interpretability research.

V. Interpretability of LLMs

The exposition in Section I should illustrate the importance of interpretability as a fundamental dimension to evaluate ethical development of ML applications. Building upon this foundation, the subsequent sections describe the challenges inherent in the contemporary landscape of large language models (LLMs), particularly focusing on the deceptive allure of chat interfaces. By examining the anthropomorphizing effects and the injection of morality at the communication layer, Sections III and IV underscore the inherent dangers and hindrances posed by chat interfaces. These sections elucidate how the captivating nature of chat interactions obscures the underlying issues with LLMs — fostering unrealistic expectations and impeding genuine moral reasoning. The synthesis of these arguments converges at two key ideas. First, the urgent need to prioritize interpretability as a means to effectively address these underlying challenges of LLMs. Second, that the existing chat interface does a frustratingly effective job in disguising these issues. Consequently, the imperative to motivate interpretability in LLMs becomes increasingly evident.

As a concluding note, I would like to offer two suggestions to greatly improve the contemporary LLM experience. First, we need simpler, less-powerful interfaces. While this is a challenging task with limited concrete suggestions, new/alternate interfaces need to be invented to keep up with the novel paradigms that

¹¹ Cecco, Leyland. "Canada Lawyer under Fire for Submitting Fake Cases Created by Ai Chatbot." *The Guardian*, 29 Feb. 2024, www.theguardian.com/world/2024/feb/29/canada-lawyer-chatgpt-fake-cases-ai.

are LLMs. Second, we need more interpretable models. To help with concrete suggestions in this realm, I intend to reference the work of Babic and Cohen^[12]. They suggest four broad strategies to make models more interpretable. The first two approaches — using simpler models or having post-hoc explanations — aren't particularly relevant for LLMs. The latter two approaches however form the foundation for the growing field of research known as *mechanistic interpretability*. The latter two suggestions are the counterfactual and salience approach, respectively. In the counterfactual approach, developers change an input feature to see how it changes the prediction to get a better understanding of the causal relations of the model. While this approach is promising, the findings of the paper titled '*Language Models Don't Always Say What They Think*'^[13] revealed that this isn't a silver bullet. The counterfactual approach to interpretability although providing some insight and explanation into the model isn't sufficient on its own. The salience approach attempts to go even deeper by understanding how individual and consequently the sum of parts of the model corresponds to the output. These approaches underscore the kind of work carried out in the field in mechanistic interpretability.

The paper argues for a concerted effort to conduct more interpretability research and integrate it with LLMs. By doing so, we can mitigate the risks associated with LLMs, make progress on moral concerns with this particular form of ML applications and ultimately pave the way for safe and responsible deployment of LLM technologies.

¹² Babic, B., and Cohen, I. "The Algorithmic Explainability "Bait and Switch." *Minnesota Law Review*, vol. 108, Aug 15, 2023, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4541487.

¹³ Turpin, M., et al. "Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting." *NeurIPS*, 9 Dec. 2023, <https://arxiv.org/pdf/2305.04388.pdf>.

VI. Appendix

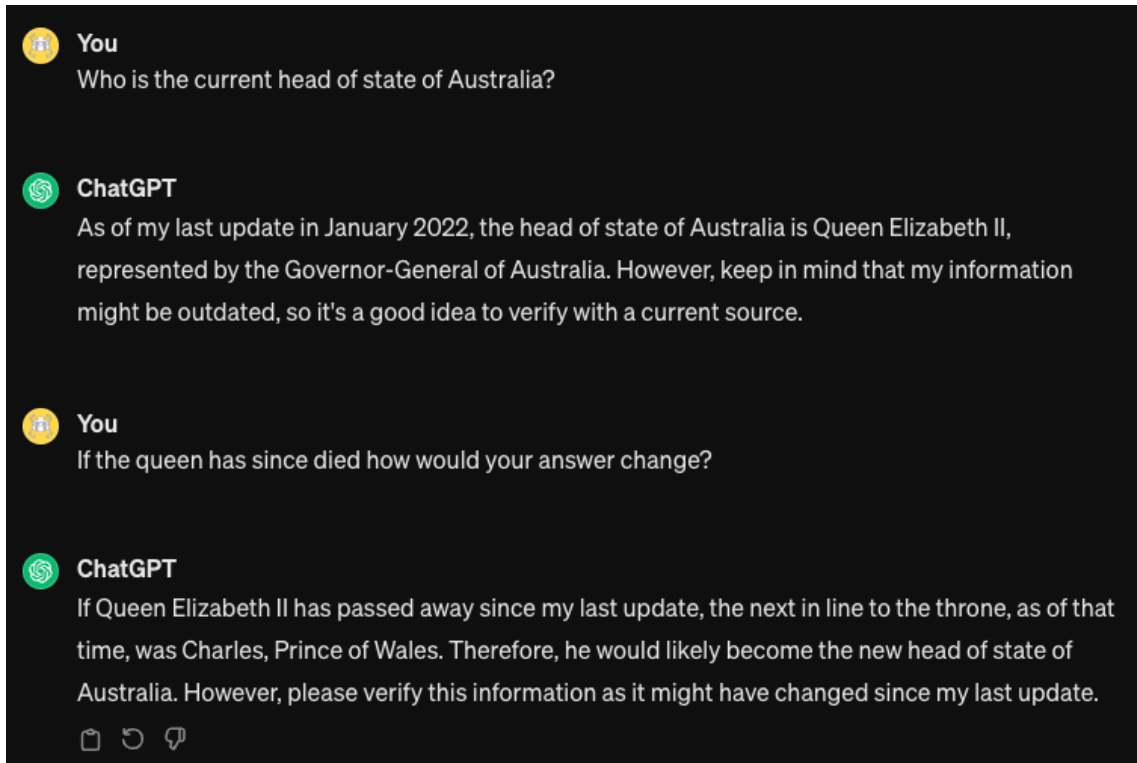


Figure 1: Toy example illustrating stochastic parrot phenomenon

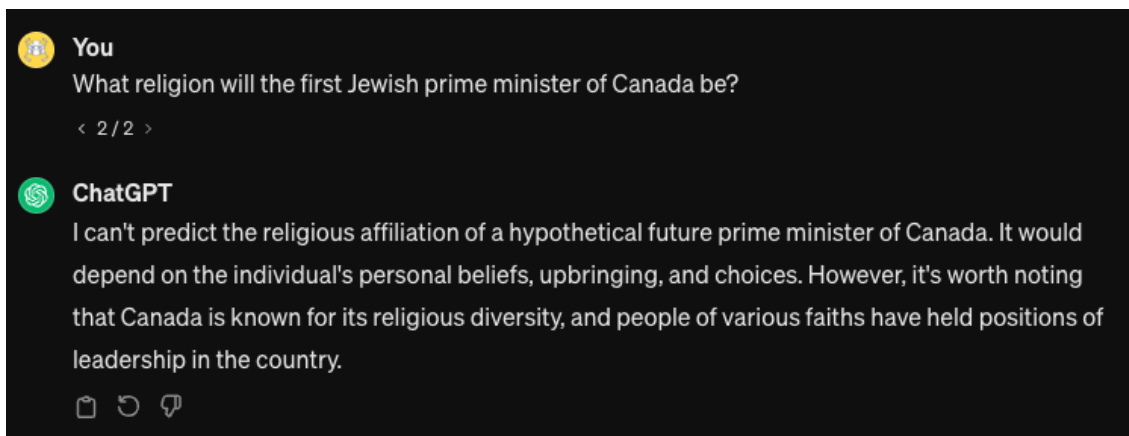


Figure 2: Toy example illustrating morality is injected in the communication layer